

Date: _____

ASSIGNMENT # 01:

Day: _____

NAME:

MUHAMMAD DANIAL.

COURSE INSTRUCTOR:

Engl. NIGHAT USMAN

COURSE TITLE:-

SE-209, Intro. to DATABASE

Reg #:

24ABSWE0021.

DUE DATE:-

05/04/2026.

Question # 1:

Explain the differences b/w DBA, DA and End User in a database environment?

Answer:

ROLES In DATABASE ENVIRONMENT:-

The primary difference b/w the DBA (Database Administrator), DA (Data Administrator) and End User in a database environment lies in the scope of their work: the DA manages the "what" (policy), the DBA manages the "how" (technical), and the End User focuses on the "result" (utility).

DIFFERENCES TABLE:-

**DATA -
ADMINISTER
(DA)**

**DATABASE
ADMINISTER
(DBA)**

**END-
USER**

PRIMARY FOCUS:-

Strategic &
Managerial:

Focuses on data
as a corporate
asset.

Technical &
Operational:

Focuses on the
physical DBMS
software.

Functional:

Focuses on using
the data to
perform tasks.

RESPONSIBILITIES:-

Establishing data
standards, policies &
business rules.

Installation, perfo-
rmance tuning,
backups etc.

Querying,
updating &
generating database reports.

TECHNICAL SKILLS:-

High business & moderate technical knowledge

Technical expertise in SQL server & DBMS internals

Varies from basic computer literacy to application-use

EXAMPLES:-

A Chief Data-
Officer deciding
that all customer
names must
follow specific
global format

A system Admin
setting up a
schedule to
backup the
database every
night at 2:00 AM

A Bank Teller
using software
interface check
a customer's
account balance.

Question # 02:-

Explain how the transition from file-based systems to database systems helped in reducing data redundancy & data consistency?

Answer:-

TABLE FOR IMPACT OF TRANSITIONS:-

Issues	File-based System Problems:	DBMS SOLUTIONS
1. Data - Redundancy	Multiple copies of the same data in different files	Centralized Storage: data is defined once and shared.
2. Data - Inconsistency	Updating one file leaves others with "stale" or wrong data.	A single update is reflected across the entire system.
3. Dependence	Program - Data Dependence (changing the structure data format requires changing the app.)	Data - Independence (the structure can change without breaking the app.)

FOR EXAMPLE:-

Student ID Uniqueness	Same Student ID can be entered multiple times, causing duplicate records and inconsistency.	Student ID is defined as unique primary key, so any duplicate entry is automatically rejected.
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